

Measuring things

In ancient times, parts of the human body were used to measure length (some systems today still use "feet"). The first weights were often based on fixed quantities of grain. These traditional units served well for thousands of years, until the rise of scientific experimentation brought the need for far more accurate methods of measuring things.

Early weights and measures

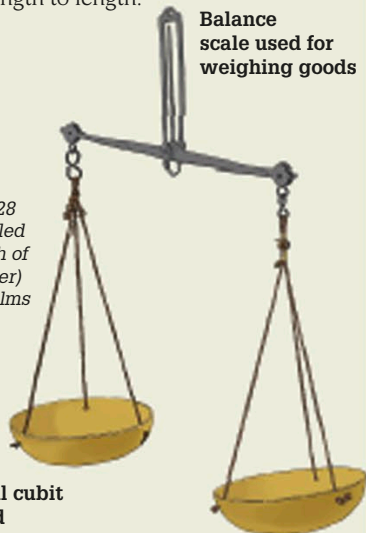
Today, we measure length, weight, and volume using international standard units. In the past, these units were local—each city or country set their own. For example, in medieval England, an inch was equivalent to three grains of barley laid length to length.

Balance scale used for weighing goods

Rod is one cubit (the length of a forearm)

Divided into 28 segments called fingers (width of a human finger) and seven palms (width of a human hand)

Egyptian royal cubit measuring rod



Thermometer with Fahrenheit scale, 1720s

Degrees of temperature

Two scales for accurately measuring temperature were invented in the 1700s—the Fahrenheit scale in 1724 and the Celsius (or centigrade) scale in 1742. Today, the Celsius scale is used in nearly every country in the world. The US still uses the Fahrenheit scale.

Measuring small distances

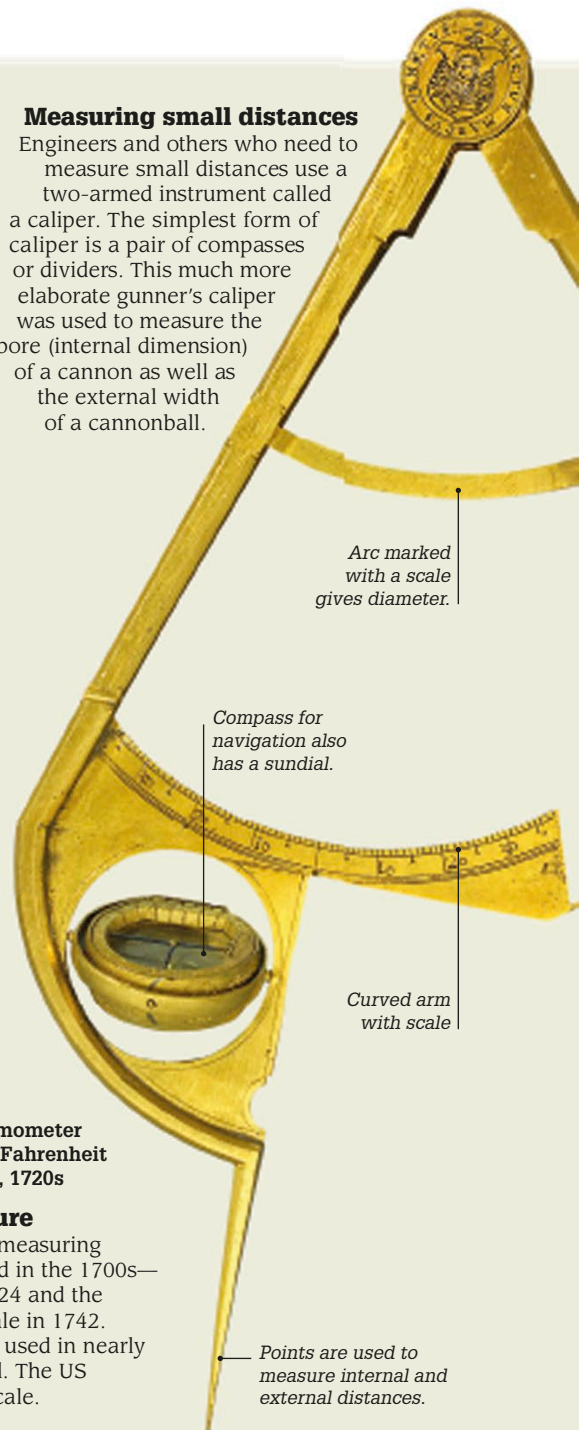
Engineers and others who need to measure small distances use a two-armed instrument called a caliper. The simplest form of caliper is a pair of compasses or dividers. This much more elaborate gunner's caliper was used to measure the bore (internal dimension) of a cannon as well as the external width of a cannonball.

Arc marked with a scale gives diameter.

Compass for navigation also has a sundial.

Curved arm with scale

Points are used to measure internal and external distances.



Key events

Egyptian royal cubit

c 3000 BCE

The royal cubit was a standard measurement of length in Ancient Egypt. It was based on the length of the forearm, from the middle fingertip to the elbow.

1631

French mathematician Paul Vernier invented a sliding scale for taking accurate measurements that are smaller than the smallest on an instrument's main scale. The vernier scale is still used today.

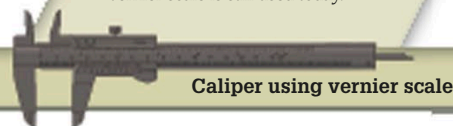
1724

Dutch physicist Gabriel Fahrenheit devised the temperature scale named after him. It has 32° as the freezing point of water, and 212° as its boiling point.



Cubit

Palm



Caliper using vernier scale